MILESTONE 5: INTERACTIVE DASHBOARD & DECISION-SUPPORT SYSTEM

1. Objective

The primary goal of Milestone 5 was to design and implement a high-fidelity, interactive dashboard in Power BI. This system functions as a Decision-Support System (DSS), enabling managers to transition from observing raw, unstructured data to making evidence-based strategic decisions. By centralizing key metrics and providing interactive exploration tools, the dashboard facilitates rapid identification of market trends and operational risks.

2. Technical Implementation

The implementation relied on a refined dataset featuring decoded categorical variables and corrected transactional metrics. The architecture prioritized low-latency interactivity and data accuracy.

2.1 Dynamic Slicing and Filtering

To ensure a highly interactive experience, several slicers were integrated into the interface:

- City Slicer: Allows users to filter all visuals by specific geographic locations (e.g., San Francisco, Los Angeles, Miami).
- Membership Category Slicer: Enables direct side-by-side comparison between Gold and Silver tier members.
- Satisfaction Level Slicer: Facilitates segment isolation based on qualitative sentiment (Satisfied, Neutral, Unsatisfied).

2.2 Core Visualizations

KPI Cards (The 'Vital Signs'):

- Total Customers: Displays the current count of the active customer base.
- Average Spend: Represents the mean financial value per transaction across the selected segment.
- Total Items: Shows the total volume of products purchased, providing an indicator of inventory throughput.

Comparative and Strategic Visuals:

- Comparative Bar Charts: Used to visualize Average Spend by City or Membership Type, highlighting the most profitable segments.
- The Strategic Scatter Plot: A multi-dimensional visual where the X-axis tracks Average Spend and the Y-axis tracks Items Corrected (Gross Volume). Color legends differentiate by Membership or Satisfaction, identifying high-frequency 'Power Users' versus churn risks.

3. Data Transformation Layer (The "Invisible" Work)

For the dashboard to provide intuitive insights, complex Data Analysis Expressions (DAX) were authored to bridge raw binary machine-readable data back into human-readable labels:

- City Decoding: Implemented via SWITCH functions to translate binary indicators (1/0) into descriptive city names.
- Membership Mapping: Transformed binary tiers into 'Gold' and 'Silver' labels, ensuring the interface remains accessible to non-technical stakeholders.

4. Decision-Support Use Cases

The system is engineered to provide actionable answers to critical business questions:

Business Question	Dashboard Action	Strategic Outcome
Identify underperforming regions	Filter by 'City'	Localized promotional targeting.
Analyze loyalty program impact	Compare 'Gold' vs 'Silver'	Refinement of membership benefits.
Prevent customer churn	Identify 'Unsatisfied' high-spenders	Immediate customer service intervention.

5. UI/UX and Human-Centered Design

Consistent color palettes (e.g., Green for Gold, Blue for Silver) were used to minimize cognitive load. The layout follows the 'F-Pattern', placing high-priority KPIs in the top-left quadrant for immediate visibility. Cross-Highlighting is enabled, ensuring that an interaction with one chart updates the entire view for synchronized analysis.

Conclusion

Milestone 5 successfully transformed abstract datasets into a navigable visual narrative. By prioritizing interactivity and clarity, the system empowers e-commerce managers to mitigate risks and capitalize on trends in real-time.